

CSA0711

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

6

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Code of Conduct:

(192525462)

I CH. ABHILASH REDDY (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: CH. ABHILASH REDDY

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5 10marks
Understanding of Scenario	20%	1.2	1.2	1.2	1.2	1.0
Identification of Key Issues	20%	1.2	1.2	1.2	1.2	1.0
Application of Concepts	25%	1.5	1.5	1.5	1.5	1.25
Decision Making	20%	1.2	1.2	1.2	1.2	1.0
Clarity of Response	15%	0.9	0.9	0.9	0.9	0.75

29/30

[Handwritten signature]

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.

2. The matching routing table entry.
3. The next-hop network/interface used to forward the packet.
4. The default route that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → 192.168.10.0/26

Finance LAN → 192.168.10.64/26

The router interfaces are configured with:

Administration Gateway → 192.168.10.1

Finance Gateway → 192.168.10.65

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario

		g of the scenario and context			
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned , logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganize d and difficult to understand

CD2-AT2

Smart Campus Subnet Allocation
Given Network.

main memory 192.168.100/0.24
total address 9n/24:

$$2^{32-24} = 256$$

Address range:

192.168.100.0 - 192.168.100.255

Departments:-

	Prefix
School of computing	125
School of Electronics	126
School of mechanical engineering	127

The Subnets are allocated from largest to smallest to avoid overlap

Subnet . masks:-

$$255.255.255.128$$

$$256 - 128 = 128$$

Therefore First Subnet is

192.168.100.0/25

Details

Address

Network address

192.168.100.0

First valid host

192.168.100.1

Last valid host

192.168.100.126

Broadcast address

192.168.100.127

$$2^{22} - 25 = 2$$

$$2^9 - 2 = 178 - 2 = 176$$

R. School of Electronics - 126

The next available address after the 125 subnets

192.168.100.128

Subnet mask:-

255.255.255.192

Block size :-

$$256 - 192 = 64$$

$$2^{32} - 26 = 2$$

$$2^6 - 2 = 64 - 2 = 62$$

C) School of Mechanical Engineering - 127

The next available address is

192.168.100.192

Subnet mask:-

255.255.255.224

$$256 - 224 = 32$$

Multinational Software company subnet allocation.

Given Network

10.10.0.0 /26

Subnet masks:-

255.255.0.0

Total number of addresses :-

$$2^{32-6} = 65,536$$

Usable address in the complete /16 network

Departments:-

$$65,536 - 2 = 65,534$$

Departments

Prefix:-

Software development

/24

Quality Assurance

/25

Human Resource

/26

Executive management

/27

The subnets are allocated frequently /16 Block

A. Software development - /24

Subnet

10.10.0.0 /24

Subnet masks:-

255.255.255.0

Details

Address.

Network address

10.10.0.0

First valid host

10.10.0.1

Last valid host

10.10.0.254

Broadcast address

10.10.0.255

$$2^{32-24} - 2 = 256 - 2 = 254$$

B. Quality Assurance - /25

The next available subnet begins at:-
10.10.1.0

Subnet:-

10.10.1.0/25

Subnet masks:- 255.255.255.128

Details

Available hosts:-

Address

$$2^{32-25} - 2$$

$$= 2 = 128 - 2 = 126$$

C. Human resources - /26

The next available address is:-

10.10.1.128

Subnet mask:-

255.255.255.192

D. Executive management

10.10.1.192

Subnet

10.10.1.192/27

Smart Hospital IPv6 Network

Given IPv6 Network

2001:0db8:abcd:0000:0000:0000:0000:164

IPv6 address contain 128 bits and are written using eight hexadecimal groups.

Compressed notation:-

Original address:

2001:0db8:abcd:0000:0000:0000:0000:0000

Remove leading zeros:

2001:db8:abcd:0:0:0:0:0

Replace the longest continuous group of zeros with ::

Compressed address:

2001:db8::abcd::164

2. Full hexadecimal representation

Compressed Form:

2001:db8:abcd::164

Expanded form:-

2001:0db8:abcd:0000:0000:0000:0000:0000:164

3. Network prefix

164 prefix address means

* First 64 bits:- Network portion.

* last 64 bits : interface (or) host position

Expanded address division

2001: db8 : abcd : 0000 : 0000 : 0000 : 0000

Therefore, the Network prefix is:-

2001: db8 : abcd : 0000 : 0000 : 0000 : 0000

It may also be written as

2001: db8 : abcd : 0000 : 0000 : 0000 : 0000

4. Total number of host address

$$128 - 64 = 64$$

$$2^{64}$$

18, 448, 744, 093, 709, 551, 616

5. Possible host address range:-

The Network begins at:-

2001: db8 : abcd : 0000 : 0000 : 0000 : 0000

The final address is:-

2001: db8 : abcd : ffff : ffff : ffff : ffff

A sample host range is

2001: db8 : abcd : 0000 : 0000 : 0000 : 0001 to 2001: db8 : abcd : ffff : ffff : ffff : ffff

ision:-

The compressed hospital network is $2001:db8:abcd::/64$. It contains 2^{64} possible interface address, making it suitable for a very large number of medical devices and sensors.

Q1

IP Routing table Analysis

Given Routing table

Routing entry

$192.168.10/24$

$192.168.20/24$

$192.168.30/24$

Destination IP

$192.168.2.45$

1. Destination Subnet:-

For destination:-

$192.168.2.45$

The network portion is:-

$192.168.2$

Therefore, the destination subnet is

$192.168.20/24$

Detail
network address

address

192.168.20

first valid host

192.168.2.1

last valid host

192.168.2.254

broadcast address

192.168.2.255

matching routing table entry

route

Does it match

192.168.1.0/24

No

192.168.0/24

yes

192.168.3.0/24

No.

Therefore, the matching route is: -

192.168.2.0/24

3. Next

Hop Network (or) Interface

192.168.2.0/24

192.168.1.0/24

Gigabit Ethernet0/0

4.

Default

route:

0.0.0.0/0

Ex:-

0.0.0.0/0 via 10.0.0.1

destinations.

10.0.0.1

The destination 192.168.20.45 belongs to the 192.168.20.0/24 network entry and therefore, the router selects the 192.168.20.1/24 as the next hop and forwards the packets through the corresponding interface.

Administration and Finance LAN Configuration
 Given information:

LAN

Network

Administrator

LAN

192.168.10.0/26

Finance

LAN

192.168.10.64/26

LAN

Administrator

Gateway.

Finance

192.168.10.1

192.168.10.65

Subnet masks for /26

Administrator

192.168.10.1

Finance

192.168.10.65

Subnet masks for 126 :-

255.255.255.192

Block size

$$256 - 192 = 64$$

$$2^6 = 64 \text{ total addresses}$$

$$64 - 2 = \boxed{62}$$

A. Administration LAN

Subnet:

192.168.10.0/26

Gateway :-

configured gateway is: ✓

192.168.10.1

Finance LAN

Subnet

192.168.10.64/26

Final LAN configuration table

LAN
Administrator
Finance

Network Host Range

192.168.10.0/26

192.168.10.1

192.168.10.64/26

192.168.10.65

- 192.168.10.126

Conclusion:-

Both LANs are correctly divided into separate
/26 Network. Each subnet supports 62 usable host
address and gateway 192.168.10.1 Final gateway 192.168.10.65